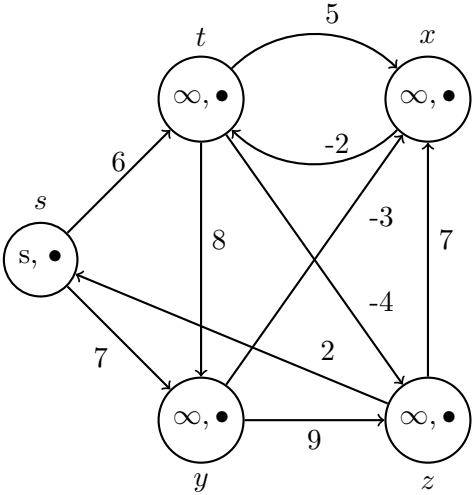
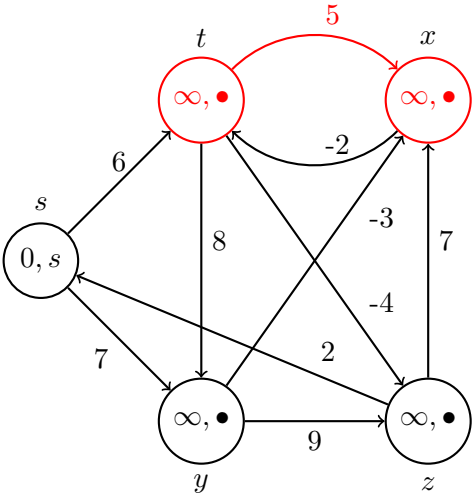
Example: Bellman-Ford



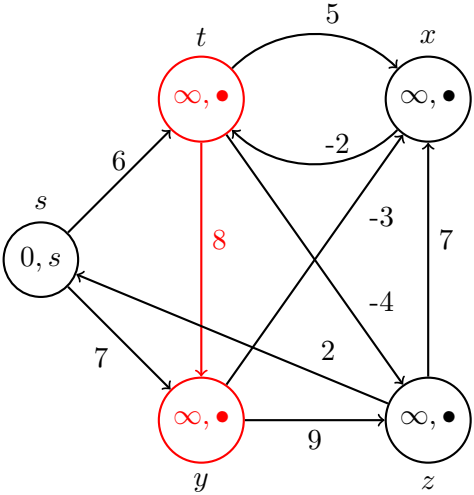
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)

Example: Bellman-Ford



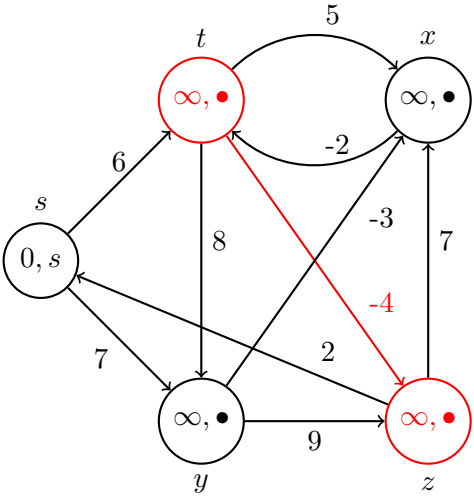
$i = 1$	(t, x)	(t, y)	(t, z)	(x, t)	(y, x)	(y, z)	(z, x)	(z, s)	(s, t)	(s, y)
	×									

Example: Bellman-Ford



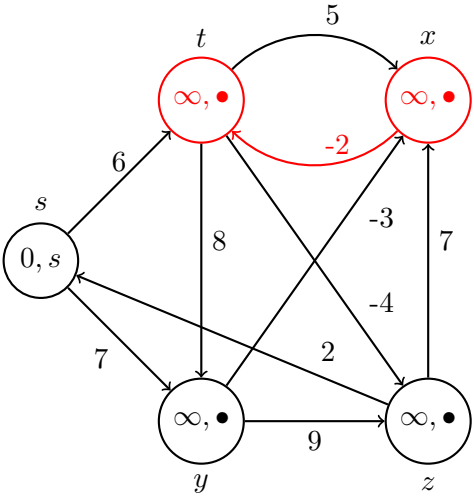
$i = 1$	(t,x)	$\boldsymbol{(t,y)}$	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	$\times$	$\times$								

Example: Bellman-Ford



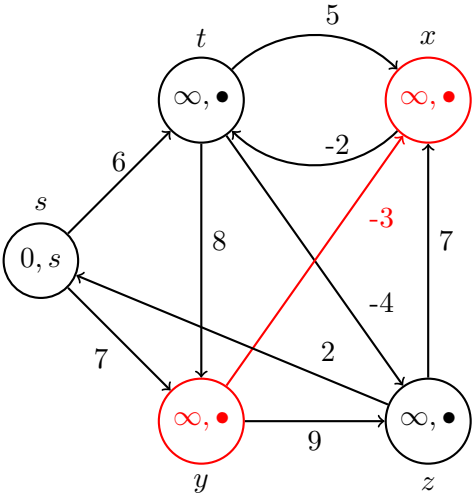
$i = 1$	(t,x)	(t,y)	$\boldsymbol{(t,z)}$	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×							

Example: Bellman-Ford



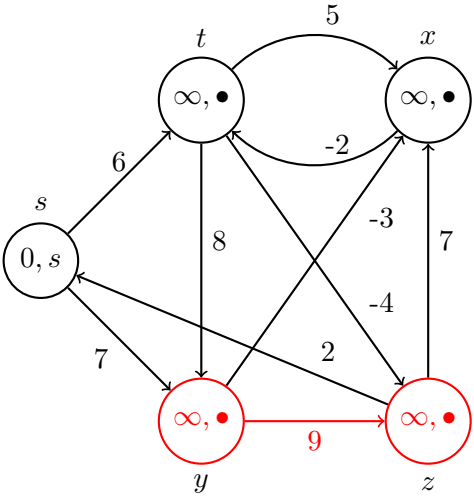
$i = 1$	(t,x)	(t,y)	(t,z)	$\boldsymbol{(x,t)}$	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	×						

Example: Bellman-Ford



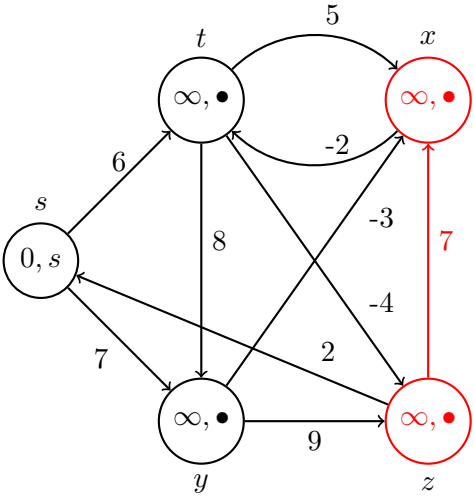
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	$\boldsymbol{(y,x)}$	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	×	×					

Example: Bellman-Ford



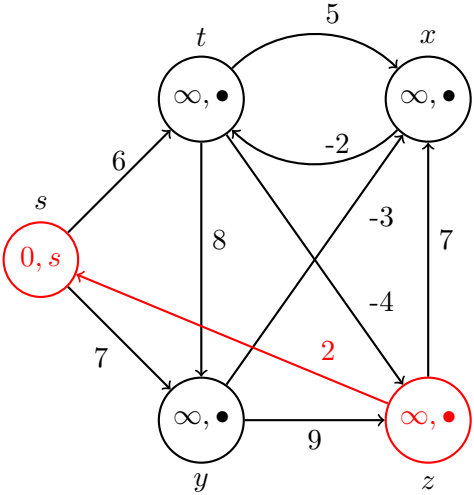
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	$\boldsymbol{(y,z)}$	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	×	×	×				

Example: Bellman-Ford



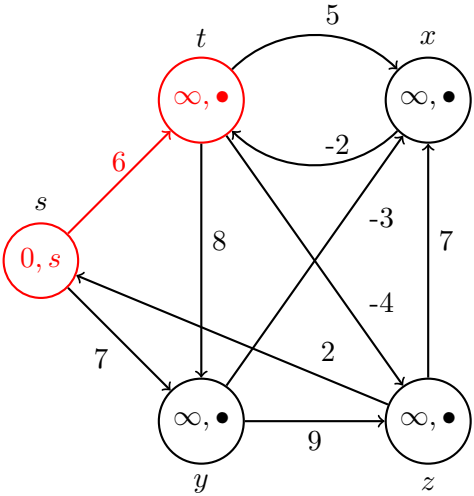
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	$\boldsymbol{(z,x)}$	(z,s)	(s,t)	(s,y)
	×	×	×	×	×	×	×			

Example: Bellman-Ford



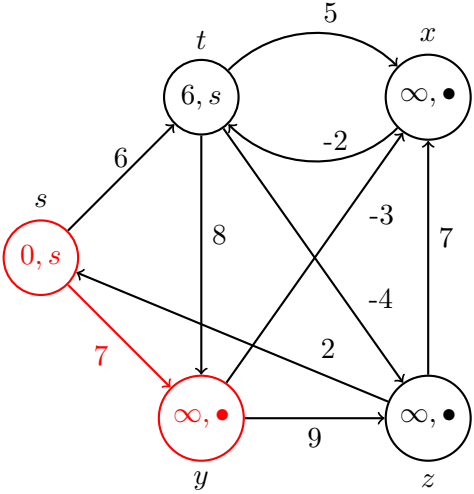
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	×	×	×	×	×		

Example: Bellman-Ford



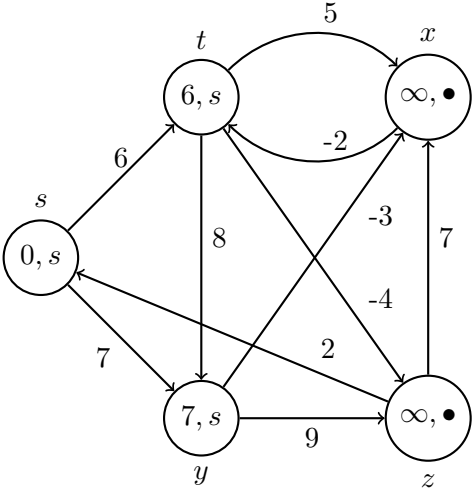
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	$(\boldsymbol{s,t})$	(s,y)
	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	

Example: Bellman-Ford



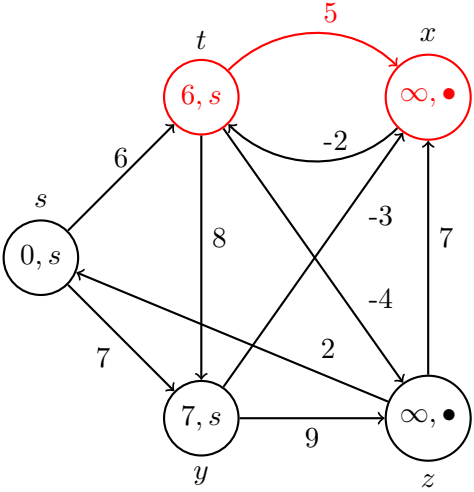
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	$\boldsymbol{(s,y)}$
	×	×	×	×	×	×	×	×	✓	✓

Example: Bellman-Ford



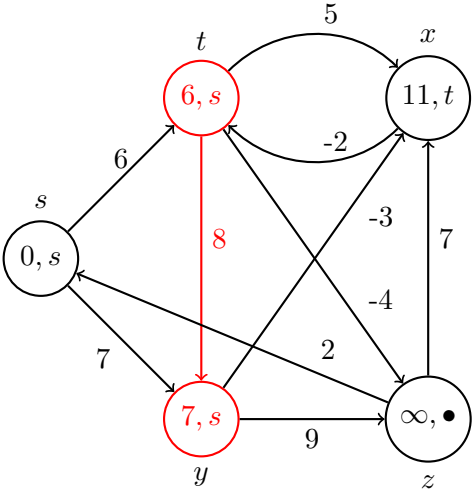
$i = 1$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	×	×	×	×	×	✓	✓

Example: Bellman-Ford



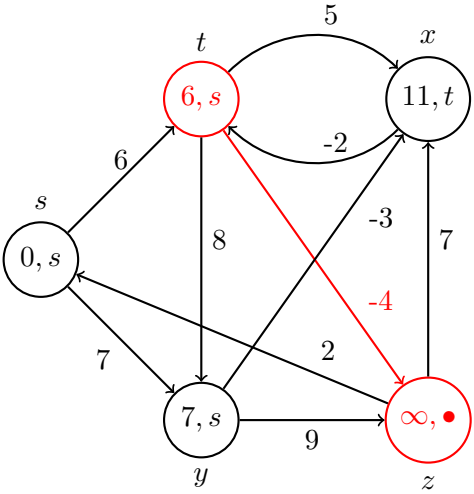
$i = 2$	(t, x)	(t, y)	(t, z)	(x, t)	(y, x)	(y, z)	(z, x)	(z, s)	(s, t)	(s, y)
	✓									

Example: Bellman-Ford



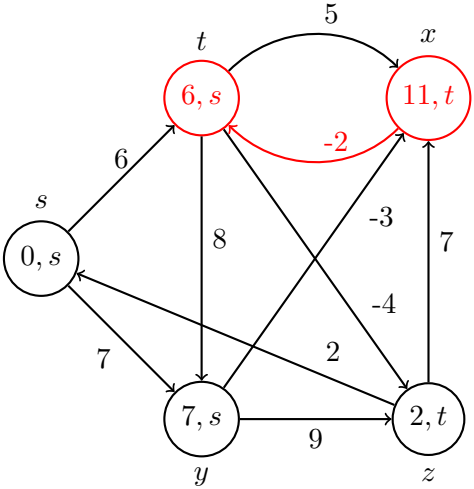
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	✓	×								

Example: Bellman-Ford



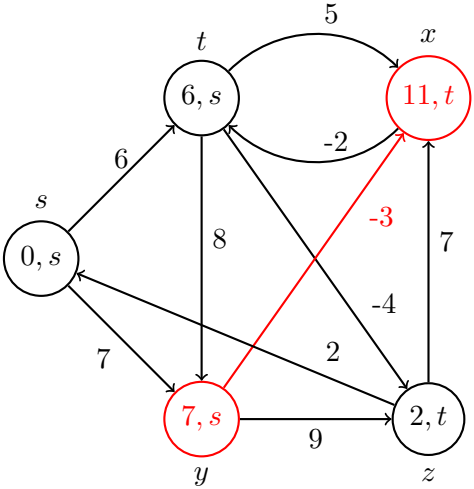
$i = 2$	(t,x)	(t,y)	$\boldsymbol{(t,z)}$	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	✓	×	✓							

Example: Bellman-Ford



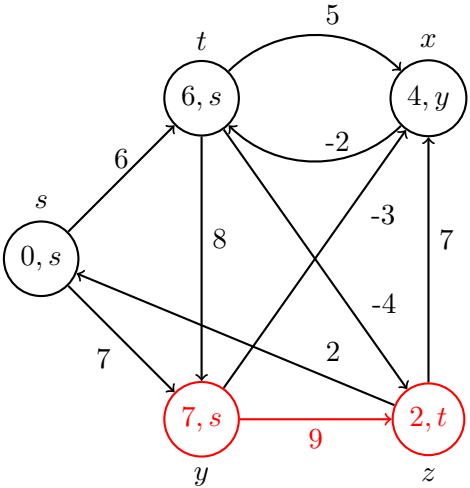
$i = 2$	(t,x)	(t,y)	(t,z)	$\boldsymbol{(x,t)}$	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	✓	×	✓	×						

Example: Bellman-Ford



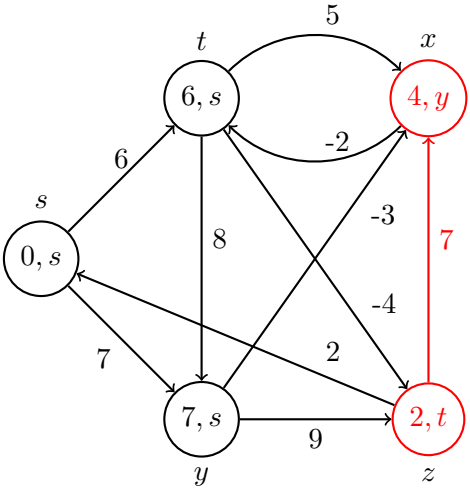
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	$\boldsymbol{(y,x)}$	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	✓	×	✓	×	✓					

Example: Bellman-Ford



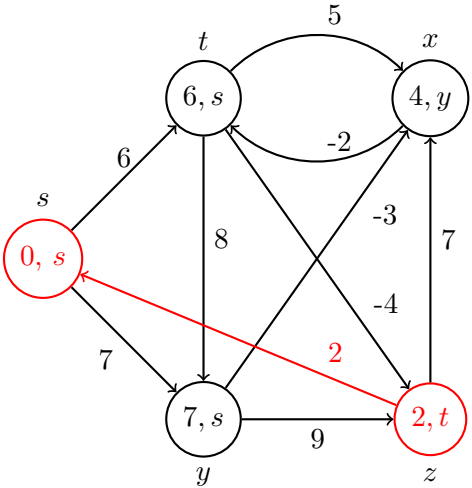
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	$\boldsymbol{(y,z)}$	(z,x)	(z,s)	(s,t)	(s,y)
	✓	×	✓	×	✓	×				

Example: Bellman-Ford



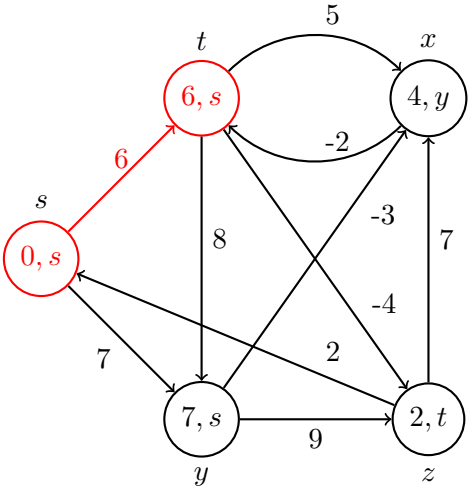
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	$\boldsymbol{(z,x)}$	(z,s)	(s,t)	(s,y)
	✓	×	✓	×	✓	×	×			

Example: Bellman-Ford



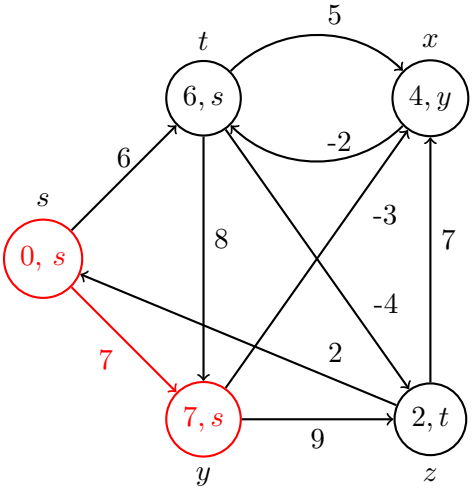
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	$\boldsymbol{(z,s)}$	(s,t)	(s,y)
	✓	×	✓	×	✓	×	×	×		

Example: Bellman-Ford



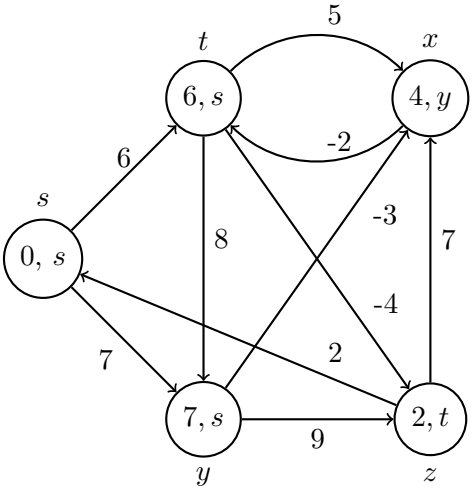
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	$(\boldsymbol{s},\boldsymbol{t})$	(s,y)
	✓	×	✓	×	✓	×	×	×	×	

Example: Bellman-Ford



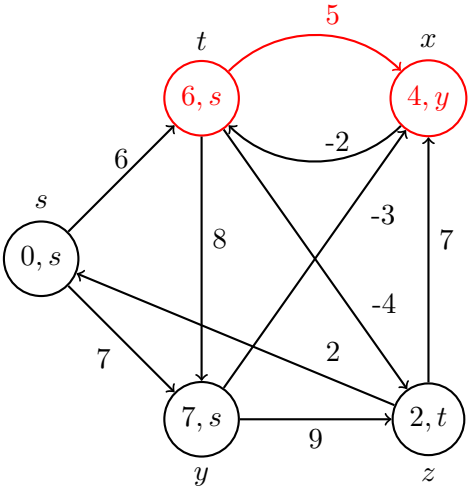
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	$\boldsymbol{(s,y)}$
	✓	×	✓	×	✓	×	×	×	×	×

Example: Bellman-Ford



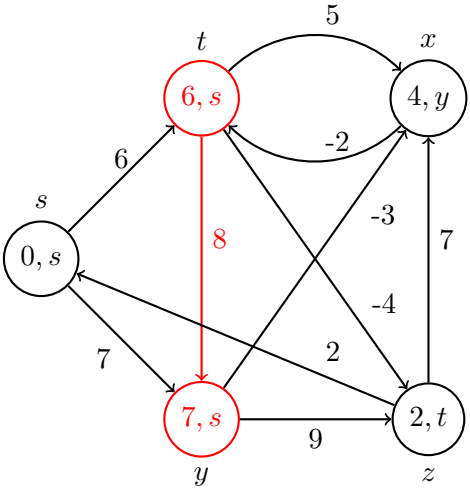
$i = 2$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	✓	×	✓	×	✓	×	×	×	×	×

Example: Bellman-Ford



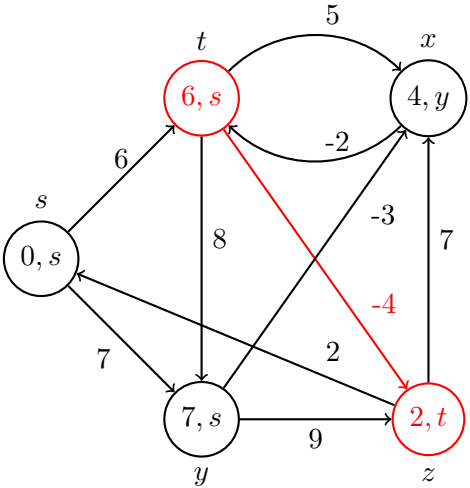
$i = 3$	(t, x)	(t, y)	(t, z)	(x, t)	(y, x)	(y, z)	(z, x)	(z, s)	(s, t)	(s, y)
	$\times$									

Example: Bellman-Ford



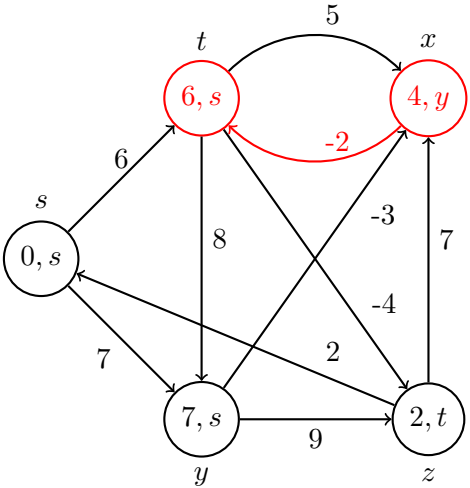
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	$\times$	$\times$								

Example: Bellman-Ford



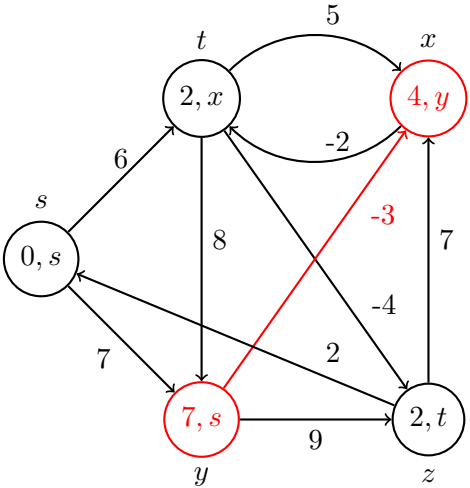
$i = 3$	(t,x)	(t,y)	$\boldsymbol{(t,z)}$	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	$\times$	$\times$	$\times$							

Example: Bellman-Ford



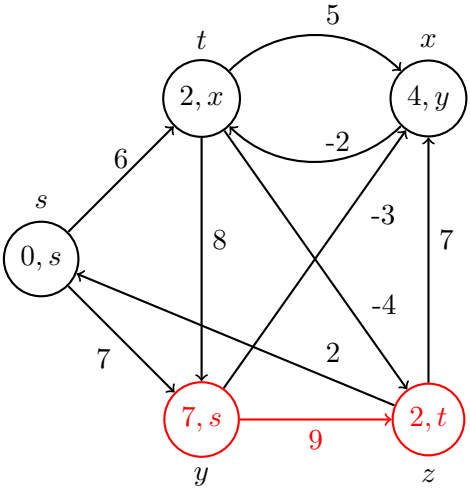
$i = 3$	(t,x)	(t,y)	(t,z)	$\boldsymbol{(x,t)}$	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	$\times$	$\times$	$\times$	$\checkmark$						

Example: Bellman-Ford



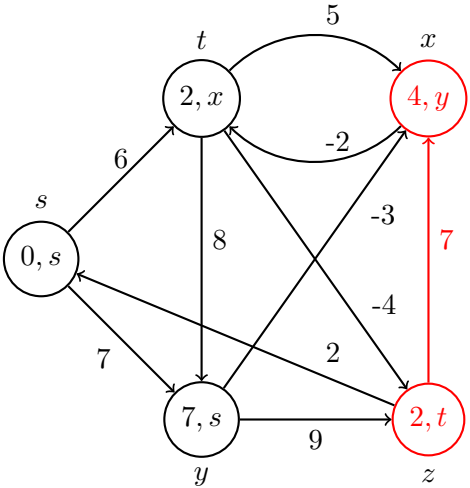
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	$\boldsymbol{(y,x)}$	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	✓	×					

Example: Bellman-Ford



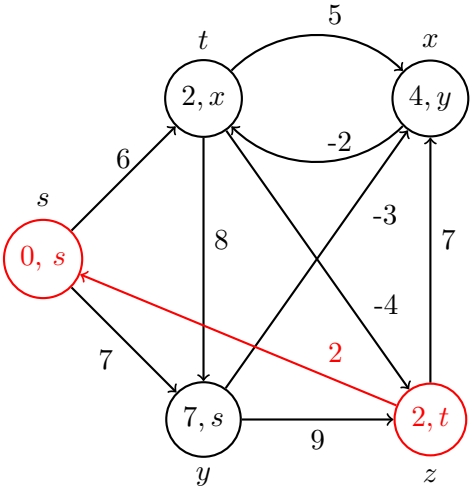
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	$\boldsymbol{(y,z)}$	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	✓	×	×				

Example: Bellman-Ford



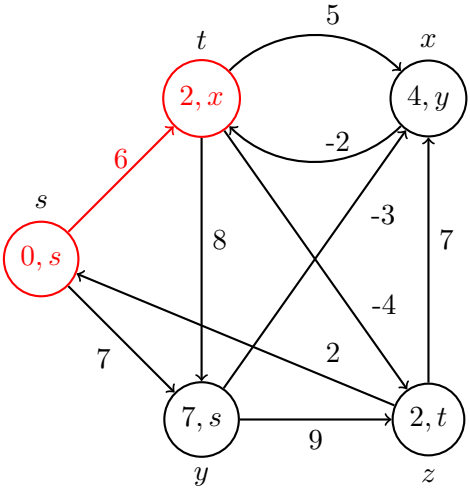
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	$\boldsymbol{(z,x)}$	(z,s)	(s,t)	(s,y)
	×	×	×	✓	×	×	×			

Example: Bellman-Ford



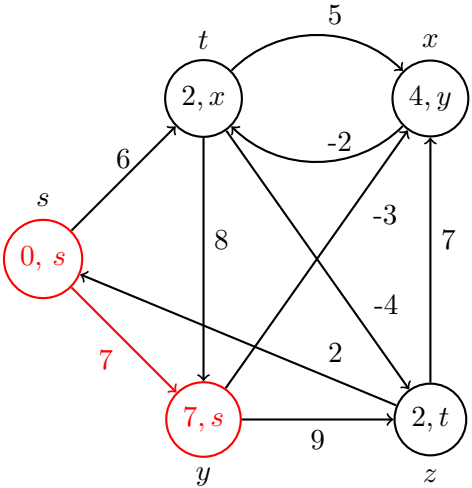
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	✓	×	×	×	×		

Example: Bellman-Ford



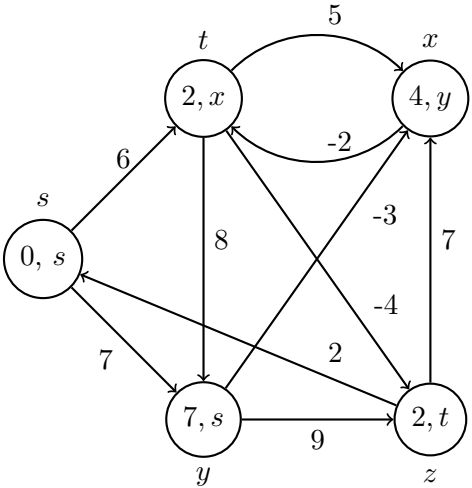
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	$(\boldsymbol{s},\boldsymbol{t})$	(s,y)
	×	×	×	✓	×	×	×	×	×	

Example: Bellman-Ford



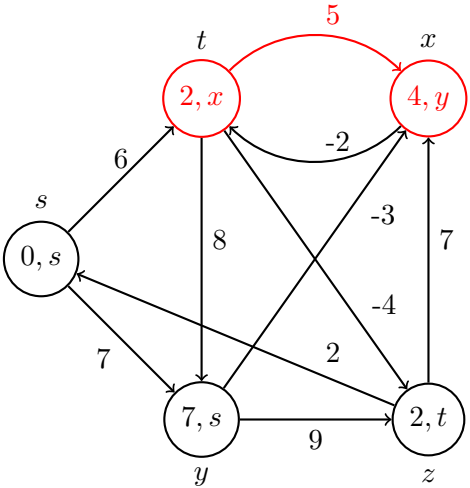
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	$\boldsymbol{(s,y)}$
	×	×	×	✓	×	×	×	×	×	×

Example: Bellman-Ford



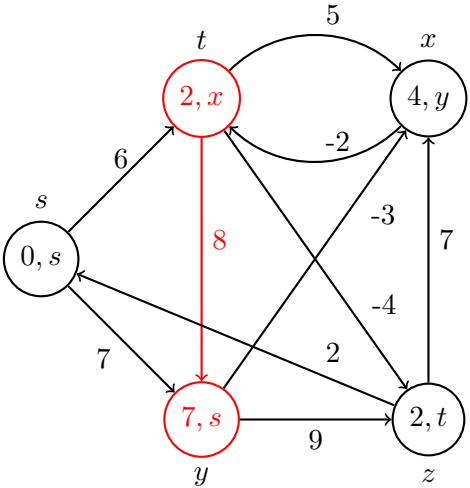
$i = 3$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	✓	×	×	×	×	×	×

Example: Bellman-Ford



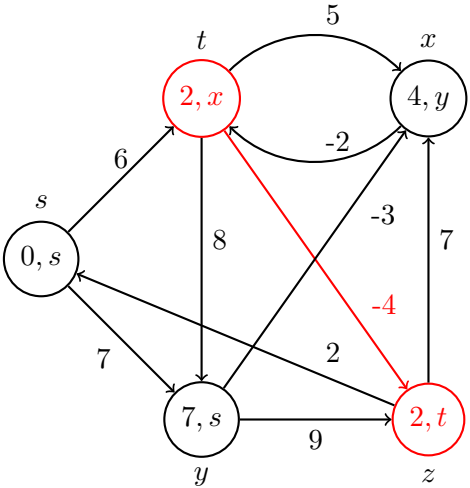
$i = 4$	(t, x)	(t, y)	(t, z)	(x, t)	(y, x)	(y, z)	(z, x)	(z, s)	(s, t)	(s, y)
	$\times$									

Example: Bellman-Ford



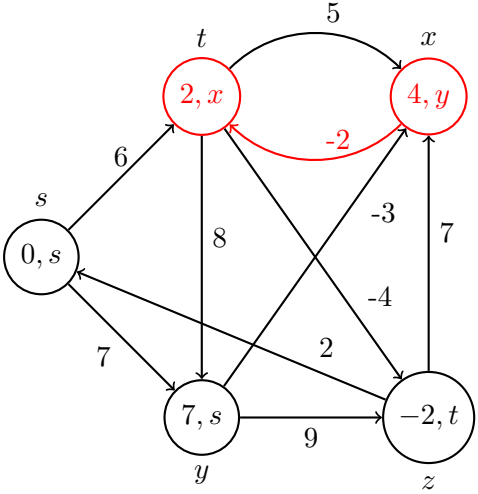
$i = 4$	(t,x)	$\boldsymbol{(t,y)}$	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	$\times$	$\times$								

Example: Bellman-Ford



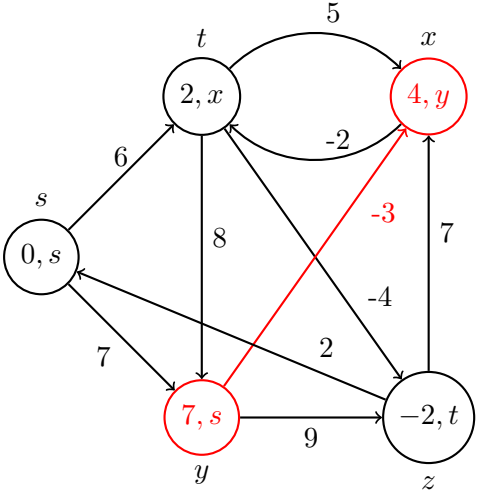
$i = 4$	(t,x)	(t,y)	$\boldsymbol{(t,z)}$	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	✓							

Example: Bellman-Ford



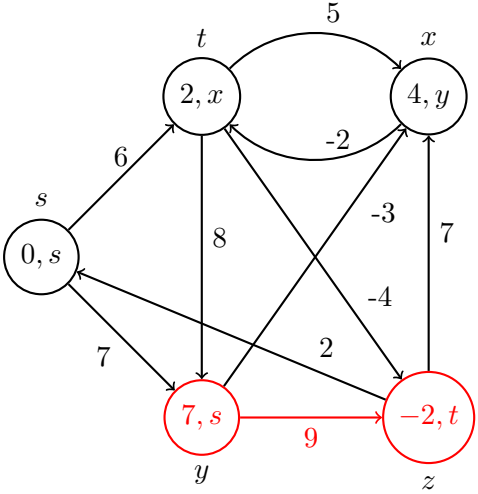
$i = 4$	(t,x)	(t,y)	(t,z)	$\boldsymbol{(x,t)}$	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	✓	×						

Example: Bellman-Ford



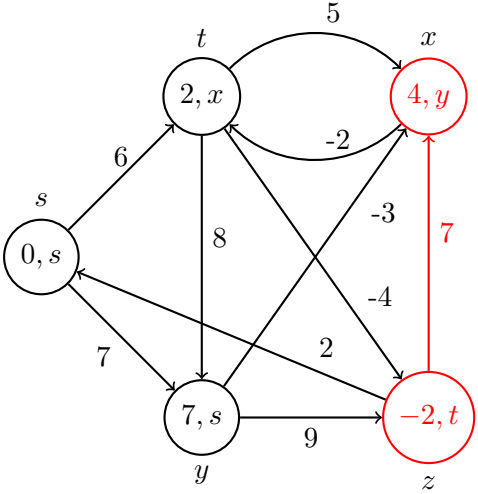
$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	$\boldsymbol{(y,x)}$	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	✓	×	×					

Example: Bellman-Ford



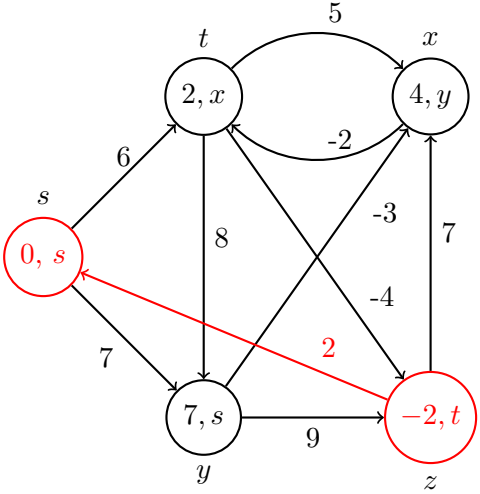
$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	$\boldsymbol{(y,z)}$	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	✓	×	×	×				

Example: Bellman-Ford



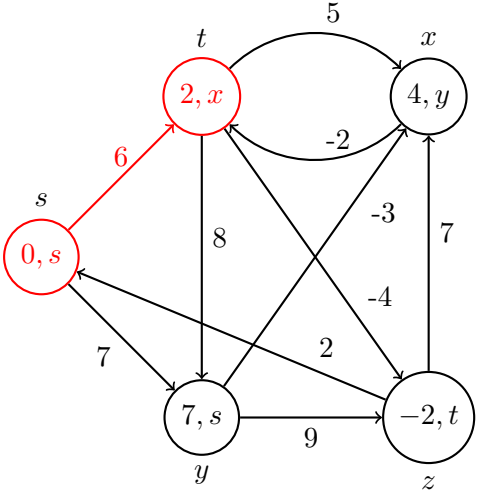
$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	✓	×	×	×	×			

Example: Bellman-Ford



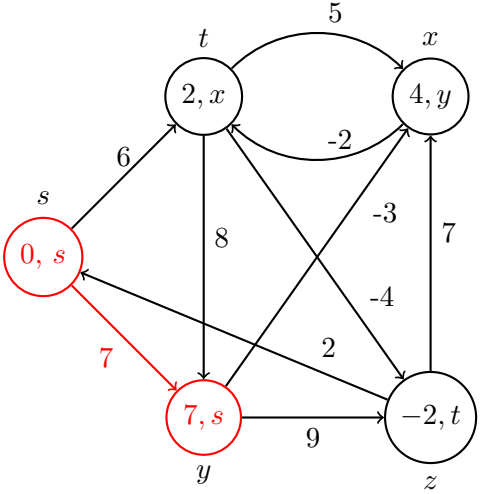
$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	✓	×	×	×	×	×		

Example: Bellman-Ford



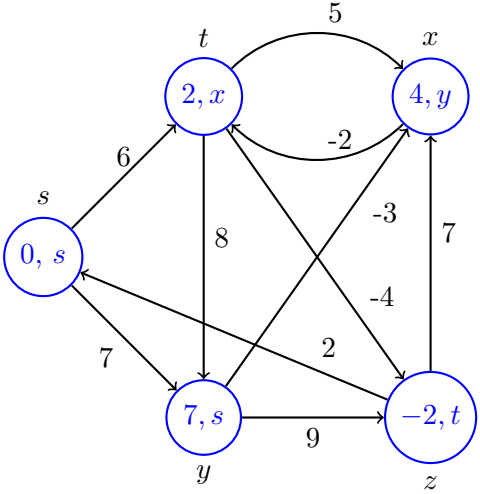
$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	$(\boldsymbol{s,t})$	(s,y)
	×	×	✓	×	×	×	×	×	×	

Example: Bellman-Ford



$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	$\boldsymbol{(s,y)}$
	×	×	✓	×	×	×	×	×	×	×

Example: Bellman-Ford



$i = 4$	(t,x)	(t,y)	(t,z)	(x,t)	(y,x)	(y,z)	(z,x)	(z,s)	(s,t)	(s,y)
	×	×	×	✓	×	×	×	×	×	×